

Solvency Field Theory

V11.1 Seed Note

The Planck-Area Accounting Lemma

From one Compton tick to boundary capacity, local gravity, and global expansion

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Status. Candidate companion lemma / V11.1 seed note. This note is not required for the V11 master argument, but sharpens one of its deepest interpretations: the Planck area is not a hidden spatial lattice cell. It is the elementary boundary-accounting scale associated with one maintenance-count event. Local curvature and global expansion are coarse-grained responses to accumulated accounting.

Abstract

Solvency Field Theory identifies persistent physical state with active maintenance. A particle with energy E is serviced at the rate E/h_P , where h_P is Planck's constant. The V11 master paper uses this count to connect local gravity and cosmic expansion: local concentrations of maintenance load produce gradients of the lag field, while the global sum of maintenance load forces boundary growth. This note isolates the Planck-scale content of that claim. The safe mathematical statement is not that there is a minimum curvature. Curvature is a coarse-grained field and can be arbitrarily small. The quantized object is the boundary-accounting increment. A single maintenance tick increments holographic capacity by one Planck-area-scale unit, conventionally written $\delta A = \xi \ell_P^2$. From this, a rest particle of mass m carries an area-demand rate $\dot{A}_{\text{tick}} = \xi Gm/(2\pi c)$. The same accounting, summed over a Hubble volume and settled against the holographic boundary, reproduces the SFT Friedmann ratio $H_{\text{SFT}}^2/H_{\text{std}}^2 = 1/(4\pi^2)$. Locally, the same maintenance density sources the weak-field lag potential through the Poisson Green's function, recovering $\psi(r) = -Gm/(c^2 r)$ for a point mass. The Planck length is therefore not an invisible lattice spacing in SFT. It is the linear scale associated with the elementary boundary-capacity unit.

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1 The question

The motivating question is simple:

Is the Planck length tied to particle-caused curvature?

The first answer should be cautious. The Planck length should not be introduced as a minimum local curvature. A curvature scalar has dimensions of inverse length squared and, in classical and semiclassical limits, can be made arbitrarily small by considering weak sources or large distances. For an isolated particle, the weak-field curvature scale at radius r behaves schematically as

$$\mathcal{R}(r) \sim \frac{Gm}{c^2 r^3}, \quad (1)$$

which has no nonzero lower bound as $r \rightarrow \infty$.

The safer SFT claim is different:

The Planck area is the elementary boundary-accounting scale. A tick does not create a minimum local curvature. A tick creates a minimum accounting increment, and curvature is the coarse-grained local response to many such increments.

This is the distinction that makes the idea mathematically usable. The quantized object is not the local curvature value. The quantized object is the capacity increment assigned to the holographic ledger.

2 Definitions and notation

The note uses the following notation.

Symbol	Meaning
h_P	Planck's constant, $h_P = 2\pi\hbar$. The subscript prevents confusion with the spatial projector.
$\hbar$	Reduced Planck constant.
ℓ_P	Planck length, $\ell_P = \sqrt{\hbar G/c^3}$.
A_P	Planck area, $A_P = \ell_P^2 = \hbar G/c^3$.
$h_{\mu\nu}$	Spatial projector orthogonal to the clock congruence, $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$.
ε	Energy density. This note uses ε to avoid confusing energy density with mass density.
p	Pressure.
ρ_{ops}	Operation-density source, $(\varepsilon + 3p)/h_P$ for a perfect fluid in the clock frame.
ξ	Area-normalization convention factor: $\delta A = \xi \ell_P^2$.

The factor ξ is deliberately kept explicit. If one adopts the SFT colloquial convention “one Planck area per tick,” then $\xi = 1$. If one tracks the Bekenstein-Hawking entropy normalization literally, one natural capacity unit carries the familiar $4\ell_P^2$ scale, up to bit versus nat conventions. The point of the lemma does not depend on this choice; the convention controls normalization, not the mechanism.

3 Planck area as capacity unit

The Planck area is

$$A_P \equiv \ell_P^2 = \frac{\hbar G}{c^3} \quad (2)$$

This expression already contains the three ingredients of the SFT mechanism:

- $\hbar$ sets the quantum phase/accounting scale.
- G converts maintenance load into geometric response.
- c sets the causal processing and handoff scale.

In standard presentations, ℓ_P is often introduced by dimensional analysis. SFT assigns it a sharper interpretation: ℓ_P^2 is the area scale of one unit of boundary capacity. The corresponding elementary accounting increment is written

$$\delta A = \sigma_A = \xi \ell_P^2 \quad (3)$$

Equivalently, one can speak in capacity units instead of area units. If the boundary capacity is

$$I(A) = \frac{A}{4\ell_P^2}, \quad (4)$$

then an area increment δA corresponds to

$$\delta I = \frac{\delta A}{4\ell_P^2} = \frac{\xi}{4}. \quad (5)$$

Different normalizations move factors between δA , δI , and the bit/nat convention. They do not change the central claim: finite boundary area means finite accounting capacity.

4 One particle: tick count becomes area demand

The SFT maintenance count for a particle with energy E is

$$\dot{N} = \frac{E}{h_P} \quad (6)$$

For a particle at rest, $E = mc^2$, so

$$\dot{N}_C = \frac{mc^2}{h_P} \quad (7)$$

This is the Compton maintenance rate. Each complete phase cycle is one count event. The elementary boundary-area demand rate of the particle is then

$$\dot{A}_{\text{tick}} = \sigma_A \dot{N}_C = \xi \ell_P^2 \frac{mc^2}{h_P}. \quad (8)$$

Now substitute

$$\ell_P^2 = \frac{\hbar G}{c^3}, \quad h_P = 2\pi\hbar. \quad (9)$$

Then

$$\dot{A}_{\text{tick}} = \xi \left(\frac{\hbar G}{c^3} \right) \left(\frac{mc^2}{2\pi\hbar} \right) \quad (10)$$

$$= \frac{\xi G m}{2\pi c}. \quad (11)$$

So the single-particle result is

$$\boxed{\dot{A}_{\text{tick}} = \frac{\xi G m}{2\pi c}} \quad (12)$$

This equation is the compact form of the Planck-area accounting insight.

Interpretation. A particle's mass sets its Compton tick rate. Its tick rate sets its boundary-area demand rate. The Planck area supplies the accounting unit, G converts energy accounting into geometry, c sets the causal processing scale, and 2π enters because SFT counts complete cycles rather than radians.

For a general energy E , the same statement is

$$\boxed{\dot{A}_E = \xi \ell_P^2 \frac{E}{h_P} = \frac{\xi G E}{2\pi c^3}} \quad (13)$$

The rest-particle equation is the special case $E = mc^2$.

5 What is quantized, and what is not

The equation above should not be read as saying that a particle places a Planck-sized chunk of curvature into nearby space at each tick. That would be the wrong object.

The tick produces a capacity-accounting increment. The local metric is the coarse-grained response of the geometry to many such increments distributed over many sources and times. In the continuum limit, the accounting increments are represented by a smooth operation-density field.

Thus the safe statement is:

$$\boxed{\text{one tick} \implies \delta A = \xi \ell_P^2} \quad (14)$$

not

$$\boxed{\text{one tick} \not\Rightarrow \text{a minimum local curvature value.}} \quad (15)$$

A curvature scalar can be arbitrarily small after coarse-graining or at large distance. The elementary object is the boundary-accounting unit, not the local curvature field.

6 Local coarse-grained limit: gravity from accumulated ticks

In a continuum region, the operation-density source is

$$\boxed{\rho_{\text{ops}} = \frac{\varepsilon + 3p}{h_P}} \quad (16)$$

for a perfect fluid measured in the local clock frame. Covariantly, this is

$$\rho_{\text{ops}} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h_{\text{P}}} \quad (17)$$

where u^μ is the clock congruence and $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ is the spatial projector. In the weak-field dust limit, $p \approx 0$, so

$$\rho_{\text{ops}} = \frac{\varepsilon}{h_{\text{P}}}. \quad (18)$$

Define the dimensionless lag potential

$$\psi \equiv \frac{\Phi_N}{c^2}, \quad (19)$$

where Φ_N is the Newtonian potential. The weak-field Poisson equation is

$$\nabla^2 \Phi_N = 4\pi G \rho_m, \quad (20)$$

with mass density $\rho_m = \varepsilon/c^2$. Therefore

$$\nabla^2 \psi = \frac{1}{c^2} \nabla^2 \Phi_N \quad (21)$$

$$= \frac{4\pi G}{c^2} \rho_m \quad (22)$$

$$= \frac{4\pi G}{c^4} \varepsilon. \quad (23)$$

Since $\varepsilon = h_{\text{P}} \rho_{\text{ops}}$ for dust,

$$\nabla^2 \psi = \frac{4\pi G h_{\text{P}}}{c^4} \rho_{\text{ops}} \quad (24)$$

This is the local weak-field accounting map: operation density sources lag curvature. For a point mass,

$$\varepsilon(\mathbf{x}) = mc^2 \delta^3(\mathbf{x}), \quad (25)$$

so

$$\nabla^2 \psi = \frac{4\pi G m}{c^2} \delta^3(\mathbf{x}). \quad (26)$$

Using

$$\nabla^2 \left(-\frac{1}{r} \right) = 4\pi \delta^3(\mathbf{x}), \quad (27)$$

one obtains

$$\psi(r) = -\frac{Gm}{c^2 r} \quad (28)$$

and hence

$$\Phi_N(r) = c^2 \psi(r) = -\frac{Gm}{r} \quad (29)$$

The gravitational field is the gradient:

$$\mathbf{g} = -\nabla\Phi_N = -\frac{Gm}{r^2}\hat{\mathbf{r}} \quad (30)$$

Local result. A local concentration of maintenance ticks coarse-grains into an operation-density source. The weak-field Green's function of that source is the usual $1/r$ lag potential. Gravity is the gradient of that accumulated maintenance-debt profile.

7 Global coarse-grained limit: expansion from the total count

The same accounting can be summed over a Hubble volume.

Let the Hubble radius be

$$R_H = \frac{c}{H}. \quad (31)$$

The Hubble volume is

$$V_H = \frac{4\pi}{3}R_H^3 = \frac{4\pi c^3}{3H^3}. \quad (32)$$

For homogeneous energy density ε , the total operation rate is

$$\dot{N}_{\text{tot}} = \frac{\varepsilon V_H}{h_P}. \quad (33)$$

Over one Hubble time,

$$t_H = \frac{1}{H}, \quad (34)$$

so the total processed count is

$$N_H = \dot{N}_{\text{tot}} t_H = \frac{\varepsilon V_H}{h_P H}. \quad (35)$$

The Hubble boundary area is

$$A_H = 4\pi R_H^2 = \frac{4\pi c^2}{H^2}. \quad (36)$$

The holographic capacity is

$$I_H = \frac{A_H}{4\ell_P^2} = \frac{\pi c^2}{\ell_P^2 H^2}. \quad (37)$$

The SFT Hubble settlement condition is

$$N_H = I_H \quad (38)$$

Substituting the expressions above gives

$$\frac{\varepsilon}{h_P H} \left(\frac{4\pi c^3}{3H^3} \right) = \frac{\pi c^2}{\ell_P^2 H^2}. \quad (39)$$

Solving for H^2 :

$$\frac{4\pi\epsilon c^3}{3h_P H^4} = \frac{\pi c^2}{\ell_P^2 H^2}, \quad (40)$$

$$\frac{4\pi\epsilon c^3}{3h_P} = \frac{\pi c^2 H^2}{\ell_P^2}, \quad (41)$$

$$H^2 = \frac{4\epsilon c \ell_P^2}{3h_P}. \quad (42)$$

Now substitute

$$\ell_P^2 = \frac{\hbar G}{c^3}, \quad h_P = 2\pi\hbar. \quad (43)$$

Then

$$H_{\text{SFT}}^2 = \frac{4\epsilon c}{3} \frac{\hbar G}{c^3} \frac{1}{2\pi\hbar} \quad (44)$$

$$= \frac{2G\epsilon}{3\pi c^2}. \quad (45)$$

The standard Friedmann equation for energy density ϵ is

$$H_{\text{std}}^2 = \frac{8\pi G}{3} \frac{\epsilon}{c^2}. \quad (46)$$

Therefore

$$\frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{\frac{2G\epsilon}{3\pi c^2}}{\frac{8\pi G\epsilon}{3c^2}} \quad (47)$$

$$= \frac{1}{4\pi^2}. \quad (48)$$

The global result is

$$\boxed{\frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{1}{4\pi^2}} \quad (49)$$

Global result. The same count that locally sources a lag gradient, when summed over the Hubble volume and settled against the holographic boundary, produces the SFT Friedmann ratio. Local gravity and global expansion are two coarse-grained faces of the same Planck-area accounting process.

8 The same mechanism in one diagram

The Planck-area accounting lemma can be summarized as three linked maps.

$$\text{one tick} \implies \delta A = \xi \ell_P^2, \quad (50)$$

$$\text{many ticks locally} \implies \nabla^2 \psi = \frac{4\pi G h_P}{c^4} \rho_{\text{ops}} \implies \text{gravity}, \quad (51)$$

$$\text{all ticks globally} \implies N_H = I_H \implies \frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{1}{4\pi^2} \implies \text{expansion}. \quad (52)$$

This is the compact mathematical form of the statement:

Gravity is what the accounting looks like when ticks concentrate. Expansion is what the accounting looks like when ticks are summed.

9 No invisible lattice

This lemma does not require a microscopic spatial lattice. It does not say that space is made of tiny cubes of side length ℓ_P . It does not say that a particle moves on pre-existing Planck-spaced rails.

Instead, it says:

$$\boxed{\ell_P^2 \text{ is the boundary capacity unit, not a bulk lattice cell.}} \quad (53)$$

The distinction matters. A lattice is extra microscopic structure. A boundary capacity unit is an accounting limit. In SFT, the smooth metric is not a continuous substance that must be quantized by adding new microscopic furniture. The smooth metric is the macroscopic geometry of an enormous accounting process.

The analogy is a digital image, but only as an analogy. A digital photograph has discrete pixels, yet the photograph is not made of a physical grid in the world it depicts. The grid is the accounting structure of the representation. Likewise, the Planck area is the accounting scale of the holographic ledger, not necessarily a literal grid embedded in space.

10 Interface with the V11 theory

This seed note does not replace any V11 theorem. It clarifies the Planck-scale interpretation of the existing V11 machinery.

V11 structure	Planck-area interpretation
Compton Maintenance Postulate	$\dot{N} = E/h_P$ is the event count.
Holographic capacity	ℓ_P^2 is the area scale of one boundary-accounting unit.
Mesoscale bridge	Many tick events coarse-grain into $\rho_{\text{ops}} = (\varepsilon + 3p)/h_P$.
Local gravity	Local tick concentration gives a lag profile; its gradient is gravity.
Global expansion	Total tick count settles against the Hubble boundary, giving expansion.
$4\pi^2$ spine	Complete-cycle counting and horizon capacity combine to produce the SFT Friedmann ratio.

The note therefore sharpens the phrase “one tick, one Planck area.” The phrase is useful, but the mathematical version is better:

$$\boxed{\text{one maintenance count event} \implies \text{one elementary boundary-capacity increment.}} \quad (54)$$

The exact area normalization depends on convention. The existence of a finite capacity increment does not.

11 Status ledger

Claim	Mathematical support	Status
Planck area is the boundary-accounting scale	$\ell_P^2 = \hbar G/c^3$ and $I = A/(4\ell_P^2)$	Stable interpretation
One rest particle has area-demand rate	$\dot{A}_{\text{tick}} = \xi \ell_P^2 m c^2 / h_P = \xi Gm/(2\pi c)$	New lemma
Minimum object is not curvature	Curvature scales such as $Gm/(c^2 r^3)$ can be arbitrarily small	Guardrail
Local tick density gives Newtonian lag	$\nabla^2 \psi = (4\pi G h_P / c^4) \rho_{\text{ops}}$, point solution $\psi = -Gm/(c^2 r)$	Weak-field support
Global tick count gives SFT Friedmann ratio	$N_H = I_H \Rightarrow H_{\text{SFT}}^2 / H_{\text{std}}^2 = 1/(4\pi^2)$	V11-consistent
No invisible lattice required	δA is a boundary capacity increment, not a bulk cell	Interpretive closure

12 Bottom line

The Planck length is not introduced in SFT as a hidden lattice spacing. The Planck area is the elementary boundary-accounting scale of the maintenance ledger. A particle tick increments that ledger. Local concentrations of such increments coarse-grain into lag gradients, which are gravity. The global sum of such increments settles against the Hubble boundary, which is expansion.

No invisible lattice. No extra microscopic furniture.
One tick, one accounting increment.
Local concentration gives gravity. Global accumulation gives expansion.

This is the same mechanism all the way down and all the way up.